ESA: Boundaries of attractors with bounded noise: boundary maps, normal hyperbolicity and singularities

Early Stage Assessment report

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Abstract

Random dynamical systems with bounded noise have attractors that are compact sets with non-empty interior. Their shape, and in particular the regularity of their boundaries, governs how they change under perturbation. The *boundary map* lifts the set-valued dynamics to a single-valued map on the unit normal bundle $\mathbb{R}^d \times \mathbb{S}^{d-1}$, where it is a conformal contactomorphism. The boundary of the attractor lifts to an invariant Legendrian submanifold. The project studies when this Legendrian submanifold is normally hyperbolic (and hence the boundary is C^2-persistent), how normal hyperbolicity is lost, and which singularities (wedges, cusps) appear on the perturbed boundary when it is. This report sets out the problem and background and then analyses the planar diagonal linear case in full. We find the fixed points of the boundary map, their spectra and eigenvectors, and the global heteroclinic structure of the invariant normal bundle. We show that normal hyperbolicity fails under any definition at the resonance $\lambda_1 = \lambda_2^2$ (Task 1 of the project), and that with uniform rates it holds if and only if $\lambda_1 > \lambda_2^2$. In that case the curvature of ∂A at its flattest points equals $(\lambda_1 - \lambda_2^2)/(\varepsilon\lambda_1)$, and it vanishes otherwise. The spectra at fixed points obey a contact pairing rule $\{\rho\} \cup \{\nu, \rho/\nu\}$, and in these terms the resonance is a collision within a conformally symplectic eigenvalue pair. Numerical computations in JAX confirm the results. We close with a plan for the nonlinear resonant perturbation $f(x) = (\lambda^2 x_1 + a x_2^2, \lambda x_2)$ and its singularities, and with two broader strands: a *linear contact algebra* of boundary maps, and the realisation of boundary maps as *Pontryagin extremal flows* of control systems.

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1 Introduction and problem statement

When the noise is *bounded*, trajectories are typically trapped in bounded invariant sets from which escape is impossible. These attractors support the ergodic stationary measures that describe the system's statistics [24], and their bifurcations are *topological*: they occur when minimal invariant sets collide with, or detach from, other invariant objects [16, 13]. Understanding the *shape* of such attractors is therefore a prerequisite for a bifurcation theory of bounded-noise systems.

We study iterations

$$x_{i+1} = f(x_i) + \xi_i, \qquad x_i \in \mathbb{R}^d, \tag{1}$$

where f is a diffeomorphism and the ξ_i are i.i.d. random variables supported on the closed ball of radius ε. The *compound* dynamics are captured by the set-valued map

$$F_{f,\varepsilon}(C) = B_\varepsilon(f(C)) = \{f(x) + y : x \in C,\ \|y\| \le \varepsilon\},$$

acting on non-empty compact sets. Attractors of (1), that is, compact sets that are attracting and minimally forward invariant, are fixed points of $F_{f,\varepsilon}$; in the linear case the fixed point is unique and globally attracting [17, 16].

The boundary map. If an attractor A is convex with smooth boundary, each boundary point x has a unique outward unit normal $n(x)$, and ∂A lifts to the *unit normal bundle* $N^1_+\partial A = \{(x, n(x)) : x \in \partial A\} \subset \mathbb{R}^d \times \mathbb{S}^{d-1}$ (Fig. 1). The *boundary map* [15, 23]

$$\beta_{f,\varepsilon}(x, n) = \left(f(x) + \varepsilon \frac{Df(x)^{-T}n}{\|Df(x)^{-T}n\|},\ \frac{Df(x)^{-T}n}{\|Df(x)^{-T}n\|}\right) \tag{2}$$

is a single-valued map that leaves $N^1_+\partial A$ invariant. It turns the set-valued problem into a classical one about an invariant submanifold of a diffeomorphism. For linear $f = L$ with spectral radius $\rho(L) < 1$, Lamb, Rasmussen and Tey [17] showed that the attractor is unique, strictly convex and C^1, and that $N^1_+\partial A$ is *globally attracting* for $\beta_{L,\varepsilon}$.

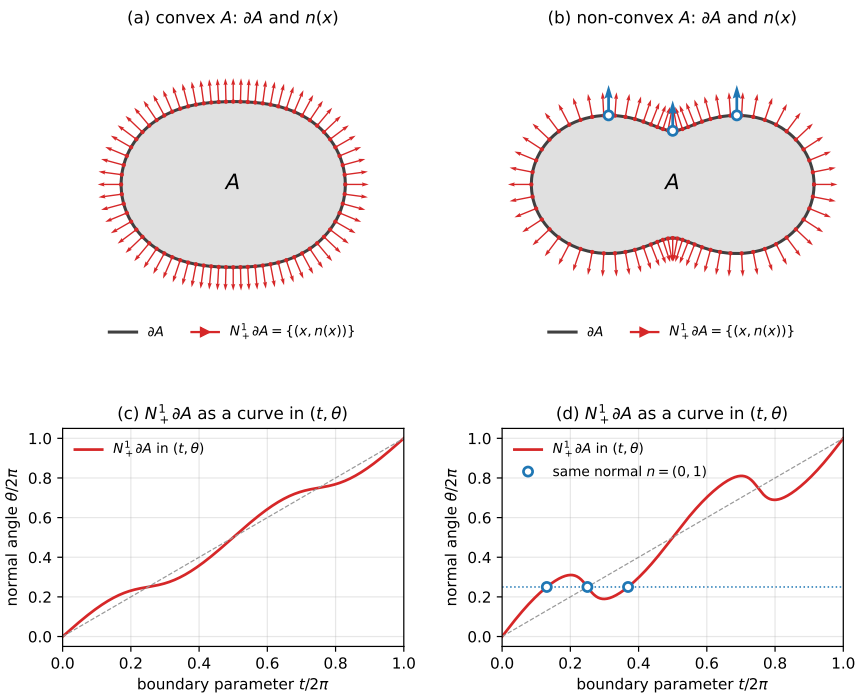


Figure 1: Outward unit normal bundles in the plane. (a),(b): a strictly convex set and a non-convex set A (shaded), with the boundary ∂A (dark grey) and the outward unit normal $n(x) \in \mathbb{S}^1$ drawn at a sample of boundary points. Everything in red is $N_+^1\partial A$: the set of all pairs $(x, n(x))$, a closed curve in $\mathbb{R}^2 \times \mathbb{S}^1$ whose front is ∂A. (c),(d): the same curve $N_+^1\partial A$ drawn in coordinates (t, θ), where t parametrises ∂A and θ is the angle of $n(x(t))$. For the convex set the angle is monotone, so $x \mapsto n(x)$ is a bijection $\partial A \to \mathbb{S}^1$ and $N_+^1\partial A$ is a graph over $\mathbb{S}^1$; this is the setting of the boundary map (2). For the non-convex set the same normal direction occurs at several boundary points, and $N_+^1\partial A$ is no longer a graph over $\mathbb{S}^1$: the three blue points in (b) all have normal $n = (0, 1)$, and in (d) they are the three intersections of $N_+^1\partial A$ with the level line $\theta = \pi/2$ (dotted).

The dual picture: support functions. Convexity is a statement about the other projection of $N^1_+\partial A$, namely $(x, n) \mapsto n$. If A is strictly convex, every local normal $n(x)$, the annihilator of $T_x\partial A$, is also a global supporting covector, the Gauss map $x \mapsto n(x)$ is a bijection $\partial A \to \mathbb{S}^{d-1}$, and ∂A is recovered from the support function $h_A(n) = \max_{y\in A}\langle n, y\rangle$ as $x(n) = \nabla h_A(n)$. When convexity fails the lift $N^1_+\partial A$ is still a smooth Legendrian, but its projection to $\mathbb{S}^{d-1}$ folds where the Gauss curvature $\det Dn_x$ vanishes. In the dual coordinates $(n, \langle n, x\rangle)$ the lift then traces a swallowtail (Fig. 2): $h_A = h_{\mathrm{conv}\,A}$ is its upper envelope, it is not differentiable in directions supported at two points, and the normals along the dented part of ∂A remain annihilators of tangent spaces without being supporting covectors. The set-valued map $F_{f,\varepsilon}$ only sees such outer envelopes, while $\beta_{f,\varepsilon}$ keeps every Legendrian lift smooth; this selection step is one mechanism by which corners can appear on perturbed boundaries.

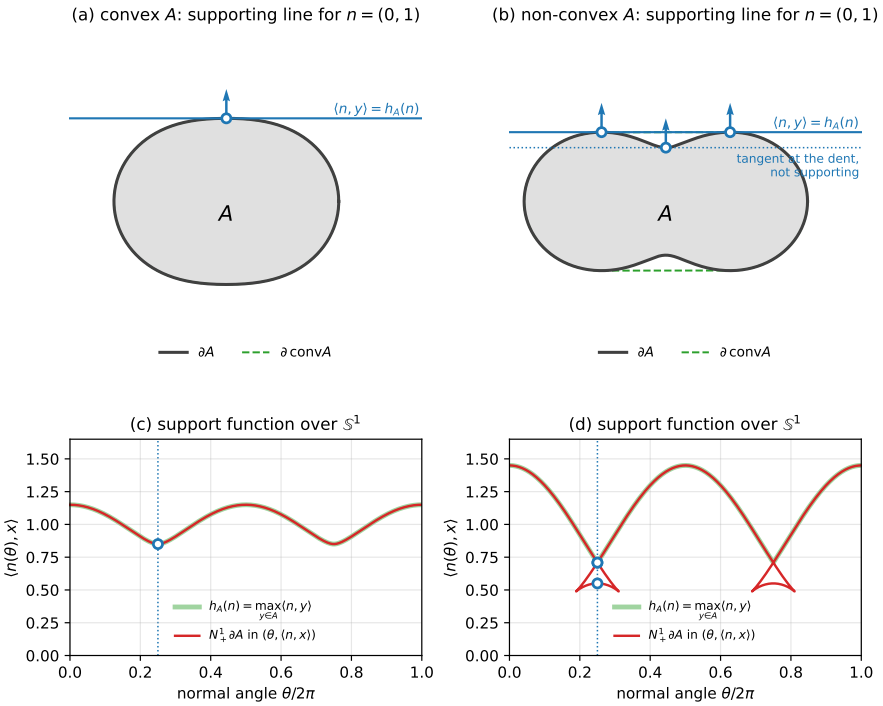


Figure 2: The Legendre-dual picture of Fig. 1. (a),(b): the sets A with their convex hull (dashed green, differing from ∂A only across the dents) and the supporting line $\langle n, y\rangle = h_A(n)$ for $n = (0, 1)$ (blue). For the convex set it touches ∂A at one point. For the non-convex set it touches at the two humps, while the tangent line at the dent point, which has the same local normal, cuts through A (dotted). (c),(d): $N^1_+\partial A$ projected to $(\theta, \langle n, x\rangle)$, the multivalued support function (red), and the true support function h_A (green). For the convex set the two coincide. For the non-convex set the red curve is a swallowtail with cusps at the zero-curvature points of ∂A; h_A is its upper envelope, with a corner at $\theta = \pi/2$ where the two hump points give the same value, and the dent point lies on the discarded lower branch.

Contact viewpoint. $\mathbb{R}^d \times \mathbb{S}^{d-1}$ carries the contact form $\alpha = n \cdot dx$. Unit normal bundles of hypersurfaces are Legendrian, and $\beta_{f,\varepsilon}$ is a (conformal) contactomorphism (Lemma 2.2). Boundary maps are thus among the simplest non-trivial contactomorphisms. Their invariant Legendrian submanifolds and the projections of these to $\mathbb{R}^d$ (*fronts*) can be expected to show, at least locally, behaviour that general contactomorphisms exhibit.

The problem. Write $f = L + \varphi$ with φ of higher order, and regard $F_{f,\varepsilon}$ as a perturbation of $F_{L,\varepsilon}$. The perturbed attractor A' is Hausdorff-close to A. By the persistence results of [15], if $\beta_{L,\varepsilon}$ is *normally hyperbolic* at $N^1_+\partial A$, then for C^2-small φ the boundary $\partial A'$ is C^2-close to ∂A (C^2-persistence). The core problem is the converse situation:

> *How is normal hyperbolicity of the boundary map lost, and which singularities appear on the boundaries of the perturbed attractors (and on their Legendrian lifts and fronts) when it is?*

In the plane, the first singularities expected on the boundary are wedges, that is, self-intersection points (A_1 in the terminology of the task sheet[1], cf. [2]). The full front may additionally carry cusps (A_2). Concretely:

Task 1. For $L = \mathrm{diag}(\lambda_1, \lambda_2)$ with $0 < \lambda_1 < \lambda_2 < 1$ and $\lambda_2^2 = \lambda_1$, show that $\beta_{L,\varepsilon}$ is not normally hyperbolic at $N^1_+\partial A$.

Task 2. At this resonance, show that there exist C^2-small perturbations $\varphi(x) = (ax_2^2, 0)$, $a \neq 0$, for which $\partial A'$ has singularities, and relate them to parameters (a, ε) for which $D\beta_{f,\varepsilon}$ has complex eigenvalues.

Task 3. Determine the type, isolatedness and stability of these singularities, via the invariant Legendrian curve $\mathcal{L}'$ and its front.

Viewing the task sheet as an instance of something more general suggests two further strands, developed in Section 4.2:

(A) Linear contact algebra. Linear symplectic algebra constrains the spectra of symplectic maps ($\nu \leftrightarrow 1/\nu$) and organises their normal forms and bifurcations (Krein collisions). The analogous question for contactomorphisms is which geometry the contact structure forces on the linearisation at fixed and periodic points, and how much of the singularity picture it predicts.

(B) Pontryagin flows and control. The set-valued map $F_{f,\varepsilon}$ is a control system $x_{i+1} = f(x_i) + u_i$, $\|u_i\| \leq \varepsilon$, whose attractor is a reachable set. The boundary map is the spherised extremal map of Pontryagin's maximum principle. This links attractor boundaries to the geometric theory of optimal control and reachable sets.

This report gives background (Section 2), presents preliminary results for the linear diagonal case, which settle Task 1 and illustrate both strands (Section 3), and describes the plan (Section 4).

2 Background and literature

2.1 Set-valued dynamics and minimal invariant sets

Equip the space $\mathcal{K}(\mathbb{R}^d)$ of non-empty compact sets with the Hausdorff metric. Lamb, Rasmussen and Rodrigues [16] developed a topological bifurcation theory for minimal invariant sets of set-valued maps such as $F_{f,\varepsilon}$. In this theory, bifurcations are discontinuous changes of minimal invariant sets in the Hausdorff metric, a phenomenon invisible to classical (deterministic or stochastic-differential) approaches. Zmarrou and Homburg [24] related such sets to bifurcations of stationary measures of random diffeomorphisms. Homburg and Young [13] studied random differential equations with bounded noise on surfaces. For discrete-time linear systems the

[1] In Arnold's notation a transverse double point of a front is the multigerm A_1^2; A_1 alone denotes a regular point.

attractor is a round ball when L is a scalar multiple of an orthogonal matrix, and Lamb, Rasmussen and Tey note, citing [13], that solid ellipsoids essentially occur only in this conformal case [17].

A recurring difficulty is that minimal invariant sets of nonlinear systems may have complicated, non-smooth boundaries and may coexist. Lamb, Rasmussen and Timperi [19, 20] classified the singularities of boundaries of ε-neighbourhoods of planar sets, which is the local geometry generated by the ball B_ε in $F_{f,\varepsilon}$. This classification is the natural local model for the boundaries considered here.

2.2 Linear maps with bounded noise

For $f = L \in GL(d, \mathbb{R})$, write $\mathcal{L}_\varepsilon = F_{L,\varepsilon}$. The following summarises [17].

Theorem 2.1 ([17])**.** *$\mathcal{L}_\varepsilon$ has a fixed point in $\mathcal{K}(\mathbb{R}^d)$ if and only if all eigenvalues of L have modulus less than 1. In that case:*

(i) the fixed point $A = \lim_{k\to\infty} \mathcal{L}_\varepsilon^k(\{0\})$ is unique and globally attracting, and is the unique attractor of (1);

(ii) A is strictly convex, and ∂A is C^1;

(iii) $N_+^1 \partial A$ is globally attracting for $\beta_{L,\varepsilon}$, and $N_+^1 \partial A = \{(x(n), n) : n \in \mathbb{S}^{d-1}\}$ with

$$x(n) = \varepsilon \sum_{k\geq 0} L^k L_\perp^{-k}(n), \qquad L_\perp(n) = L^{-T} n / \|L^{-T} n\|;$$

(iv) $\beta_{L,\varepsilon}|_{N_+^1 \partial A}$ is topologically conjugate to $L_\perp$ on $\mathbb{S}^{d-1}$.

Existence follows from the Banach fixed point theorem in an adapted norm. Convexity comes from the nested convex sets $\mathcal{L}_\varepsilon^k(0)$. Smoothness follows because every boundary point lies on the boundary of an ε-ball contained in A, so supporting hyperplanes are unique [21]. Two questions are left open in [17]: the *precise* smoothness of ∂A (beyond C^1), and extensions to non-spherical noise. The paper also notes that C^2 smoothness is what is needed to apply the persistence theory of [15]. The present project addresses the first question (Section 3.4) and then goes beyond the linear setting.

2.3 The boundary map and its geometry

The boundary map (2) factorises as

$$\beta_{f,\varepsilon} = \phi_\varepsilon \circ P \circ h_f, \qquad \mathbb{R}^d \times \mathbb{S}^{d-1} \xrightarrow{h_f} \mathbb{R}^d \times (\mathbb{R}^d \setminus \{0\}) \xrightarrow{P} \mathbb{R}^d \times \mathbb{S}^{d-1} \xrightarrow{\phi_\varepsilon} \mathbb{R}^d \times \mathbb{S}^{d-1}, \quad (3)$$

with $h_f(x, n) = (f(x), Df(x)^{-T} n)$ (the dynamics), $P(x, n) = (x, n/\|n\|)$ (renormalisation) and $\phi_\varepsilon(x, n) = (x + \varepsilon n, n)$ (the noise). Each factor has a geometric meaning, as follows.

Normals transform by Df^{-T}. A normal n to a hypersurface S at x is characterised by what it annihilates: $\langle n, u\rangle = 0$ for all $u \in T_x S$. Pushing a curve $\gamma \subset S$ with $\gamma(0) = x$, $\gamma'(0) = u$ forward gives $(f \circ \gamma)'(0) = Df(x)u \in T_{f(x)} f(S)$. So the new normal n' must satisfy $\langle n', Df(x)u\rangle = \langle Df(x)^T n', u\rangle = 0$ for all u, giving $n' \propto Df(x)^{-T} n$. Normals are covectors, which explains the inverse transpose in (2).

The ε-translation preserves normals. If Γ is a hypersurface with unit normal field n and $\Gamma_\varepsilon = \{x + \varepsilon n(x)\}$ is its outer equidistant, then differentiating $\Psi(x) = x + \varepsilon n(x)$ along a curve gives $D\Psi_x u = (I + \varepsilon Dn_x)u$. Differentiating $\|n\|^2 = 1$ shows $Dn_x u \perp n(x)$, hence $T_{\Psi(x)}\Gamma_\varepsilon = T_x \Gamma$,

and $n(x)$ is also the normal of Γ_ε at $\Psi(x)$. This justifies ϕ_ε.

Renormalisation. $D\pi_v = \frac{1}{\|v\|}\big(I - \frac{vv^T}{\|v\|^2}\big) = \frac{1}{\|v\|}\Pi_{v^\perp}$ for $\pi(v) = v/\|v\|$. It kills the radial direction and scales the rest by $1/\|v\|$.

Lemma 2.2 (conformal contact property)**.** *Let $\alpha = n \cdot dx$ on $\mathbb{R}^d \times \mathbb{S}^{d-1}$. Then $\beta_{f,\varepsilon}^* \alpha = \|Df(x)^{-T} n\|^{-1} \alpha$. In particular, $\beta_{f,\varepsilon}$ is a contactomorphism, and $N_+^1 \partial A$ is an invariant Legendrian submanifold whose front (projection $(x, n) \mapsto x$) is ∂A.*

Proof. Write $\beta_{f,\varepsilon}(x, n) = (f(x) + \varepsilon n', n')$ with $n' = Df(x)^{-T} n / \|Df(x)^{-T} n\|$. Then $\beta^* \alpha = n' \cdot (Df(x)\, dx + \varepsilon\, dn') = (Df(x)^T n') \cdot dx + \varepsilon\, n' \cdot dn'$. Since $\|n'\| = 1$, we have $n' \cdot dn' = 0$, and $Df(x)^T n' = n/\|Df(x)^{-T} n\|$. On $N_+^1 \partial A$, α vanishes because dx is tangent to ∂A and hence orthogonal to n. □

The contact property has a symplectic origin. Replace unit normals by unnormalised covectors (costates) $p \in \mathbb{R}^d \setminus \{0\}$.

2.4 Hyperbolicity, invariant manifolds and persistence

A fixed point p of a C^1 map T is hyperbolic if $\sigma(DT(p)) \cap \{|z| = 1\} = \emptyset$. The stable and unstable manifold theorem [14] then gives C^1 manifolds $W^{s,u}(p)$ with $T_p W^{s,u} = E^{s,u}$. More generally, a compact invariant manifold N is *normally hyperbolic* (attracting) if there is a DT-invariant splitting $T_N M = TN \oplus E^s$ with constants $C > 0$ and $0 < a < b$ such that, for all $k \geq 0$,

$$\|DT^k|_{E^s}\| \leq C a^k, \qquad m\big(DT^k|_{TN}\big) \geq C^{-1} b^k, \tag{4}$$

where m is the minimum norm. In words, normal contraction dominates tangential contraction. Fenichel [10] and Hirsch–Pugh–Shub [12] proved that normally hyperbolic manifolds persist under C^1-small perturbations, with smoothness controlled by the ratio of the rates (r-normal hyperbolicity: $a < b^r$ gives C^r persistence). The persistence paper [15] adapts this to boundary maps. Normal hyperbolicity of $\beta_{L,\varepsilon}$ at $N_+^1 \partial A$ yields C^2-persistence of attractor boundaries under nonlinear perturbations. Losing normal hyperbolicity is therefore a natural (though not the only) route to singular boundaries.

2.5 Legendrian fronts and their singularities

Legendrian submanifolds of $(\mathbb{R}^d \times \mathbb{S}^{d-1}, \alpha)$ project to *fronts* in $\mathbb{R}^d$, which may be singular even when the Legendrian is smooth [2, 9, 11]. For planar fronts, the generic stable singularities are transverse self-intersections (A_1^2, "wedges" when they lie on the boundary of a set) and semicubical cusps (A_2), where the Legendrian curve becomes tangent to the fibre $\{x\} \times \mathbb{S}^1$. These are exactly the singularities that appear on boundaries of ε-neighbourhoods of planar sets [19, 20].

2.6 Symplectic linear algebra and control: the two strands

For a symplectic linear map, $\nu \in \sigma$ implies $1/\nu \in \sigma$, and eigenvalues on the unit circle can leave it only through collisions of eigenvalues of opposite Krein signature [6]. For *conformally* symplectic maps ($\omega \mapsto \rho\, \omega$), the pairing becomes $\nu \leftrightarrow \rho/\nu$, and Wojtkowski and Liverani [8] showed the corresponding pairing of Lyapunov exponents. Conformally symplectic systems model dissipative mechanics and admit a KAM theory [4]. A contactomorphism with $T^*\alpha = \rho\alpha$ acts conformally symplectically on the contact planes $\ker \alpha$ with respect to $d\alpha$, which is the starting point of strand (A).

On the control side, the link between bounded-noise systems and control theory is classical: minimal invariant sets correspond to *control sets* [5], and the geometric theory of optimal control

describes reachable-set boundaries through Pontryagin extremals and their singularities [7, 3]. **??** identifies the boundary map with these extremals, which is the starting point of strand (B).

2.7 Assessment and gaps

The linear theory [17] establishes existence, convexity and C^1 smoothness, and the persistence theory [15] gives C^2-persistence under normal hyperbolicity. Missing are: (a) a characterisation of *when* $\beta_{L,\varepsilon}$ is normally hyperbolic in terms of $\sigma(L)$, together with the exact regularity of ∂A; (b) a description of what happens to $\partial A'$ when normal hyperbolicity fails; and (c) a connection between the dynamics of $\beta_{f,\varepsilon}$ (spectra at fixed points, invariant manifolds) and the singularity theory of fronts. Question (a) is answered for diagonal planar L below. Questions (b) and (c) form the main thesis programme.

The resonant example of Task 2 deserves comment. For a contraction with eigenvalues $\lambda_1 = \lambda_2^2$, the monomial $x_2^2 e_1$ is a *resonant* term: by Poincaré–Sternberg theory [22] it cannot be removed by a smooth change of coordinates. The perturbation $\varphi = (ax_2^2, 0)$ is therefore not a coordinate artefact. It is the first genuinely nonlinear term compatible with the resonance.

3 Preliminary results: the planar diagonal case

Throughout this section $d = 2$, $f = L = \operatorname{diag}(\lambda_1, \lambda_2)$ with $0 < \lambda_1 < \lambda_2 < 1$, and we write

$$\mu := \frac{\lambda_1}{\lambda_2} \in (0,1), \qquad \beta := \beta_{L,\varepsilon}, \qquad \beta(x,n) = \big(Lx + \varepsilon L_\perp(n),\ L_\perp(n)\big), \quad L_\perp(n) = \frac{L^{-1}n}{\|L^{-1}n\|}.$$

We use coordinates (x_1, x_2, θ) on $\mathbb{R}^2 \times \mathbb{S}^1$ with $n = (\cos\theta, \sin\theta)$, and the frame $\{\partial_{x_1}, \partial_{x_2}, \partial_\theta\}$, where ∂_θ corresponds to $(-\sin\theta, \cos\theta)$.

3.1 Skew-product structure and fixed points

For linear f the normal component of β does not depend on x:

$$\beta(x,n) = \big(\beta_1(x,n), \beta_2(n)\big), \qquad \beta_1(x,n) = Lx + \varepsilon\beta_2(n), \qquad \beta_2 = L_\perp.$$

So β is a *skew product*: the normal dynamics on $\mathbb{S}^1$ are autonomous and drive the position dynamics. This triangular structure is what makes the linear case solvable. It breaks for nonlinear f because $Df(x)$, and hence the normal update, then depends on x.

Proposition 3.1 (fixed points)**.** *β has exactly four fixed points:*

$$p_\pm = \Big(\big(\pm\tfrac{\varepsilon}{1-\lambda_1}, 0\big),\ \pm e_1\Big), \qquad q_\pm = \Big(\big(0, \pm\tfrac{\varepsilon}{1-\lambda_2}\big),\ \pm e_2\Big).$$

Proof. $\beta(x,n) = (x,n)$ is equivalent to (II) $L_\perp(n) = n$ and (I) $Lx + \varepsilon n = x$. Equation (II) says that $L^{-1}n = \gamma n$ with $\gamma = \|L^{-1}n\| > 0$, i.e. n is a unit eigenvector of L (eigenvalue $\gamma^{-1} > 0$). Since $\lambda_1 \neq \lambda_2$, we have $\det(L - \gamma^{-1}I) = 0$ only for $\gamma^{-1} \in \{\lambda_1, \lambda_2\}$, giving $n \in \{\pm e_1, \pm e_2\}$. All four survive because both eigenvalues are positive. Then (I) gives $x = \varepsilon(I - L)^{-1}n$, which is well defined since $1 \notin \sigma(L)$. □

The second equation has $d - 1$ unknowns and is solved first; the first then determines x. This mirrors the triangular structure. Geometrically, $p_\pm$ and $q_\pm$ are the points of ∂A where the outward normal is an eigenvector of L. Since $\frac{1}{1-\lambda_1} < \frac{1}{1-\lambda_2}$, the attractor is elongated along e_2, the weakly contracting direction (Fig. 3(a)).

3.2 Linearisation

By the chain rule applied to (3), in ambient coordinates on $\mathbb{R}^2 \times \mathbb{R}^2$ we get

$$\begin{aligned} J_\beta(x,n) = J_{\phi_\varepsilon} J_P J_{h_L} &= \begin{pmatrix} I & \varepsilon I \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ 0 & \frac{1}{\|v\|}\Pi_{v^\perp} \end{pmatrix} \begin{pmatrix} L & 0 \\ 0 & L^{-T} \end{pmatrix} \\ &= \begin{pmatrix} L & \varepsilon\, DL_\perp \\ 0 & DL_\perp \end{pmatrix}, \qquad DL_\perp \coloneqq \tfrac{1}{\|v\|}\Pi_{v^\perp} L^{-T}, \end{aligned} \tag{5}$$

with $v = L^{-T}n$. Two consequences follow.

Lemma 3.2. *(i) $\sigma(J_\beta) = \sigma(L) \cup \sigma(DL_\perp)$, because the characteristic polynomial of a block upper-triangular matrix is the product of those of its diagonal blocks.*

(ii) The subspace $V = \mathbb{R}^2 \oplus 0$ of pure position displacements is invariant, with $J_\beta(w,0) = (Lw, 0)$ at every point. The "old position" has no influence on the "new direction".

In (5) the block $DL_\perp$ always has a zero eigenvalue in the radial direction. This is an artefact of working in $\mathbb{R}^2 \supset \mathbb{S}^1$, and restricting to $T\mathbb{S}^1 = n^\perp$ removes it. Doing so gives a 3×3 Jacobian in the frame $\{\partial_{x_1}, \partial_{x_2}, \partial_\theta\}$. In angle coordinates the induced circle map is

$$\tan\theta' = \mu \tan\theta, \qquad g'(\theta) \coloneqq \frac{d\theta'}{d\theta} = \frac{\mu}{\cos^2\theta + \mu^2 \sin^2\theta} \in [\mu, \mu^{-1}]. \tag{6}$$

Proposition 3.3 (spectra and eigenvectors)**.** *In the frame $\{\partial_{x_1}, \partial_{x_2}, \partial_\theta\}$,*

$$J_\beta(p_+) = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & \varepsilon\mu \\ 0 & 0 & \mu \end{pmatrix}, \qquad J_\beta(q_+) = \begin{pmatrix} \lambda_1 & 0 & -\varepsilon\mu^{-1} \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \mu^{-1} \end{pmatrix}.$$

Hence $\sigma(J_\beta(p_\pm)) = \{\lambda_1, \lambda_2, \mu\} \subset (0,1)$, so $p_\pm$ are sinks, and $\sigma(J_\beta(q_\pm)) = \{\lambda_1, \lambda_2, \mu^{-1}\}$, so $q_\pm$ are saddles with one-dimensional unstable manifolds. The eigenvectors are given in Table 1, with

$$R = \frac{\varepsilon\lambda_1}{\lambda_1 - \lambda_2^2} \quad (\lambda_1 \neq \lambda_2^2), \qquad \tilde{R} = \frac{\varepsilon\lambda_2}{\lambda_2 - \lambda_1^2} > 0.$$

p_+ (sink)			q_+ (saddle)		
eigenvalue	eigenvector	type	eigenvalue	eigenvector	type
λ_1	$\partial_{x_1} = (1,0,0)$	normal	λ_1	∂_{x_1}	normal
λ_2	$\partial_{x_2} = (0,1,0)$	normal	λ_2	∂_{x_2}	normal
μ	$(0, R, 1)$	tangent*	μ^{-1}	$(-\tilde{R}, 0, 1)$	tangent

Table 1: Spectral data at p_+ and q_+; "tangent" means tangent to $N_+^1 \partial A$ (*when $\lambda_1 > \lambda_2^2$, see Section 3.4). At p_- and q_- the eigenvalues coincide, and the θ-eigenvectors are $(0, -R, 1)$ and $(\tilde{R}, 0, 1)$ (Remark 3.4).

Proof. At p_+, $\theta = 0$ and $g'(0) = \mu$ by (6). Since $\partial_\theta n' = (-\sin\theta', \cos\theta') = e_2$ at $\theta' = 0$, the x-component $Lx + \varepsilon n'$ has θ-derivative $\varepsilon\mu e_2$. At q_+, $\theta = \pi/2$: by (6), $g'(\pi/2) = \mu^{-1}$ and $\partial_\theta n' = -e_1$, which gives the matrices. For the eigenvectors, at p_+ with eigenvalue μ we need $(\lambda_2 - \mu)w_2 + \varepsilon\mu w_3 = 0$; taking $w_3 = 1$ gives $w_2 = \varepsilon\mu/(\mu - \lambda_2) = R$. At q_+ with eigenvalue μ^{-1} we need $(\lambda_1 - \mu^{-1})w_1 - \varepsilon\mu^{-1} = 0$, so $w_1 = -\tilde{R}$. The remaining eigenvectors are immediate since L is diagonal. □

Remark 3.4 (the $\pm$ symmetry). The involution $\sigma(x,n) = (-x,-n)$ satisfies $\beta_{f,\varepsilon} \circ \sigma = \sigma \circ \beta_{f,\varepsilon}$ whenever f is odd, $f(-x) = -f(x)$: then $Df(-x) = Df(x)$ and $\beta_{f,\varepsilon}(-x,-n) = -\beta_{f,\varepsilon}(x,n)$. This is why $p_+ \leftrightarrow p_-$ and $q_+ \leftrightarrow q_-$ share spectra, with $D\sigma = \mathrm{diag}(-1,-1,1)$ mapping the eigenvectors as in Table 1. The symmetry *breaks* for the Task 2 map $f_a(x) = (\lambda^2 x_1 + a x_2^2, \lambda x_2)$, which is even in x_2. Since $f_a(-x) = -f_{-a}(x)$, σ instead conjugates $\beta_{f_a,\varepsilon}$ to $\beta_{f_{-a},\varepsilon}$, so the sides of A' near p_+ and p_- swap roles when a changes sign. The reflection $(x_1, x_2, \theta) \mapsto (x_1, -x_2, -\theta)$ remains a symmetry for every a, because f_a commutes with $\mathrm{diag}(1,-1)$.

3.3 Global structure of the invariant normal bundle

By (6) the circle map $L_\perp$ has attracting fixed points $\theta \in \{0, \pi\}$ (derivative μ) and repelling fixed points $\theta = \pm\pi/2$ (derivative μ^{-1}). Every other orbit moves monotonically from $\pm e_2$ towards $\pm e_1$.

Proposition 3.5. *$N^1_+\partial A = \{p_+, p_-\} \cup W^u(q_+) \cup W^u(q_-)$. Each $W^u(q_\pm)$ consists of $q_\pm$ and two branches, which are heteroclinic orbits to p_+ and p_-. In particular, the unstable set $\Sigma = \bigcup_{p \in \mathrm{Fix}(\beta)} W^u(p)$ equals $N^1_+\partial A$, and all recurrent dynamics of $\beta|_{N^1_+\partial A}$ is at the four fixed points.*

Proof. [Draft note: [added] The notes pose the identity as a question ("what makes this true?"); this is a proof sketch using Theorem 2.1.] By Theorem 2.1(iii),(iv), $N^1_+\partial A$ is a compact invariant curve, the graph of $n \mapsto x(n)$ over $\mathbb{S}^1$, on which β is conjugate to $L_\perp$. Let $z = (x(n), n) \in N^1_+\partial A$ with $n \notin \{\pm e_1\}$. Then n lies in an open arc bounded by some $\pm e_2$, and $L_\perp^{-k}(n) \to \pm e_2$, so $\beta^{-k}(z) \to q_\pm$ and $z \in W^u(q_\pm)$. Conversely, $N^1_+\partial A$ is a β-invariant curve through the saddle $q_\pm$ consisting near $q_\pm$ of points whose backward orbits stay close to $q_\pm$, so it coincides locally with the one-dimensional local unstable manifold. Then $W^u(q_\pm) = \bigcup_k \beta^k(W^u_{\mathrm{loc}}) \subset N^1_+\partial A$ by invariance. Finally, $p_\pm \notin W^u(q_\pm)$ because they are fixed. □

This yields an algorithm for computing $N^1_+\partial A$ (and later the Legendrian curve $\mathcal{L}'$ for nonlinear f, where no series formula is available) by *fundamental-domain iteration*. The steps are:

1. Take the unstable eigenvalue $\kappa = \mu^{-1}$ and eigenvector v^u of $J_\beta(q)$.
2. Seed N points on the fundamental domain $\{q + t v^u : t \in [\delta/\kappa, \delta]\}$ with $\delta \approx 10^{-7}$. This segment is $O(\delta^2)$-close to $W^u(q)$ because $T_q W^u = E^u$.
3. Iterate under β. The images $\beta^k(S)$ tile $W^u(q)$, and the initial error is contracted transversally (rates λ_1, λ_2) while the arc grows (rate κ).

3.4 Normal hyperbolicity and the resonance $\lambda_1 = \lambda_2^2$ (Task 1)

[Draft note: [added] The ingredients (spectra, R, the invariant subspace V) are all in the notes. The interpretation below was worked out while drafting, so check it carefully.]

Which direction is tangent to $N^1_+\partial A$ at the sink p_+? Orbits approach p_+ along the weakest contracting direction (Remark 3.9). If $\lambda_1 > \lambda_2^2$ then $\mu > \lambda_2$, so $N^1_+\partial A$ is a C^1 graph over $\mathbb{S}^1$ with $T_{p_+} N^1_+\partial A = \mathrm{span}\{(0, R, 1)\}$ (tangential rate μ), and the normal directions $\partial_{x_1}, \partial_{x_2}$ contract at rates λ_1, λ_2. If $\lambda_1 < \lambda_2^2$ then $T_{p_+} N^1_+\partial A = \mathrm{span}\{\partial_{x_2}\}$ (tangential rate λ_2), because $x'(\theta) \to \infty$ (Proposition 3.8). At the resonance $\lambda_1 = \lambda_2^2$ the eigenvector $(0, R, 1)$ *degenerates*: $\mu = \lambda_2$ is a double eigenvalue, $J_\beta(p_+)$ has a 2×2 Jordan block (confirmed numerically), and again $T_{p_+} N^1_+\partial A = \mathrm{span}\{\partial_{x_2}\}$.

Proposition 3.6. *(i) At the resonance $\lambda_1 = \lambda_2^2$, $N^1_+\partial A$ admits no $D\beta$-invariant complement at p_+. Hence $\beta_{L,\varepsilon}$ is not normally hyperbolic at $N^1_+\partial A$ in any sense.* (Task 1.)

(ii) With normal hyperbolicity in the uniform-rate form (4), $\beta_{L,\varepsilon}$ is normally hyperbolic at $N^1_+\partial A$ if and only if $\lambda_1 > \lambda_2^2$.

Proof sketch. (i) $J_\beta(p_+) = \lambda_1 \oplus J_2(\lambda_2)$, with a Jordan block on $\operatorname{span}\{\partial_{x_2}, \partial_\theta\}$. Its invariant 2-planes are $\operatorname{span}\{\partial_{x_1}, \partial_{x_2}\}$ and $\operatorname{span}\{\partial_{x_2}, \partial_\theta\}$, and both contain $T_{p_+}N^1_+\partial A = \operatorname{span}\{\partial_{x_2}\}$.

(ii) *Sufficiency.* Take $E^s = V = \mathbb{R}^2 \oplus 0$. By Lemma 3.2(ii) it is invariant along all of $N^1_+\partial A$, with $D\beta^k|_{E^s} = L^k$, so $\|D\beta^k|_{E^s}\| = \lambda_2^k$. It is complementary to $TN^1_+\partial A$ because $N^1_+\partial A$ is a C^1 graph over $\mathbb{S}^1$: the derivative series for $x'(\theta)$ converges uniformly when $\lambda_1 > \lambda_2^2$. The θ-coordinate then gives a norm on $TN^1_+\partial A$ uniformly equivalent to the ambient one, and in it $m(D\beta^k|_{TN^1_+\partial A}) \geq \prod_j g'(\theta_j) \geq \mu^k$ by (6). So (4) holds with $a = \lambda_2 < b = \mu$. *Necessity.* Let $\lambda_1 \leq \lambda_2^2$. At the saddle q_+ the eigenvalues $\lambda_1, \lambda_2, \mu^{-1}$ are distinct and $T_{q_+}N^1_+\partial A = E^u$, so the only invariant complement is V. On V, $\|D\beta^k\| = \lambda_2^k$, which forces $a \geq \lambda_2$. At p_+, $T_{p_+}N^1_+\partial A = \operatorname{span}\{\partial_{x_2}\}$ is contracted at rate λ_2, which forces $b \leq \lambda_2$. Hence no constants $a < b$ exist. □

Remark 3.7. The necessity argument is *global*: it compares the normal rate at $q_\pm$ with the tangential rate at $p_\pm$. For $\lambda_1 < \lambda_2^2$ the *local* domination condition at p_+ holds again (normal rates λ_1, μ versus tangential λ_2). Numerical checks along the heteroclinic orbit $q_+ \to p_+$ suggest that, under the relative (pointwise) definition of Hirsch–Pugh–Shub [12], normal hyperbolicity may in fact hold in this regime, with a splitting that rotates from V at q_+ to $\operatorname{span}\{\partial_{x_1}, (0, R, 1)\}$ at p_+. Only the resonance fails under every definition.

[Draft note: Check which definition the persistence paper [15] uses. The referee check suggests it uses global constants, which matches (4) and (ii).]

The geometric meaning is the curvature of ∂A at its flattest points. Parametrise ∂A by the normal angle, $\theta \mapsto x(\theta) = x(n(\theta))$. Then $x'(\theta)$ is tangent to ∂A, and $|x'(\theta)|$ is the radius of curvature. When $N^1_+\partial A$ is a C^1 graph, its tangent vector $(x'(\theta), 1)$ at a fixed point is the θ-eigenvector. So $\tilde{R}$ *is the radius of curvature of* ∂A *at* $q_\pm$*, and, when* $\lambda_1 > \lambda_2^2$*,* R *is the radius of curvature at* $p_\pm$*.* (For $\lambda_1 < \lambda_2^2$ we have $R < 0$, and $(0, R, 1)$ is a strong-stable direction transverse to $N^1_+\partial A$.)

Proposition 3.8 (curvature at $p_\pm$ and $q_\pm$)**.** *The curvature of* ∂A *at* $q_\pm$ *is* $(\lambda_2 - \lambda_1^2)/(\varepsilon\lambda_2) > 0$*. At* $p_\pm$*:*

(i) if $\lambda_1 > \lambda_2^2$*, the curvature is* $\kappa(p_\pm) = \dfrac{\lambda_1 - \lambda_2^2}{\varepsilon\lambda_1} > 0$*;*

(ii) if $\lambda_1 \leq \lambda_2^2$*, the radius of curvature diverges as* $\theta \to 0$ *and* $\kappa(p_\pm) = 0$*. Numerically, the divergence is logarithmic,* $|x'(\theta)| \sim \frac{\varepsilon \log(1/|\theta|)}{\log(\lambda_2/\lambda_1)}$*, at the resonance, and a power law* $|x'(\theta)| \asymp |\theta|^{-\alpha}$ *with* $\alpha = \log(\lambda_2^2/\lambda_1)/\log(\lambda_2/\lambda_1) \in (0, 1)$ *when* $\lambda_1 < \lambda_2^2$*.*

Proof of (i) and divergence in (ii). Since L is symmetric, $L_\perp^{-k}(n) = L^k n/\|L^k n\|$, so by Theorem 2.1(iii)

$$x_2(\theta) = \varepsilon \sum_{k \geq 0} \frac{\lambda_2^{2k} \sin\theta}{\sqrt{\lambda_1^{2k}\cos^2\theta + \lambda_2^{2k}\sin^2\theta}}.$$

Differentiating termwise at $\theta = 0$ gives $x_2'(0) = \varepsilon \sum_k (\lambda_2^2/\lambda_1)^k$, which equals $\varepsilon\lambda_1/(\lambda_1 - \lambda_2^2) = R$ if $\lambda_2^2 < \lambda_1$ (uniform convergence of the derivative series near $\theta = 0$). Otherwise $x_2'(\theta) \to \infty$ as $\theta \to 0$ by Fatou's lemma, since the terms are positive. Also $x_1'(0) = 0$ by symmetry. The computation at $q_\pm$ is identical with the roles of the indices exchanged, and it always converges since $\lambda_1^2 < \lambda_2$. The asymptotic rates in (ii) follow heuristically by counting the terms with $\mu^{-k}|\theta| \lesssim 1$ and are confirmed numerically in Fig. 4. □

Some checks and consequences:

- For $\lambda_1 = \lambda_2 = \lambda$ the formulas reduce to $\kappa = (1 - \lambda)/\varepsilon$, the curvature of the disc $A = B_{\varepsilon/(1-\lambda)}(0)$.

- In all regimes the curvature $1/|x'(\theta)|$ is continuous, tending to 0 at $p_\pm$ when $\lambda_1 \le \lambda_2^2$. So ∂A is C^2 in the diagonal planar case. This is consistent with the task sheet's statement that ∂A is C^2, which goes beyond the C^1 proved in [17].

- Higher regularity is subtler. At the resonance the curvature has no Hölder modulus at $p_\pm$. For $\lambda_1 > \lambda_2^2$, numerically $(R - |x'(\theta)|)/R \asymp |\theta|^\gamma$ with $\gamma = \log(\lambda_1/\lambda_2^2)/\log(\lambda_2/\lambda_1)$. Note that $\gamma < 1$ exactly when $\lambda_1^2 < \lambda_2^3$, i.e. when the 2-normal hyperbolicity condition $\lambda_2 < \mu^2$ fails at $p_\pm$. This links r-normal hyperbolicity to the precise smoothness question left open in [17].

[Draft note: The "not better than" statements are conjectures from the asymptotics. They need proofs, and are natural first lemmas for the thesis.]

Remark 3.9 (a dynamical reading). At p_+, orbits approach along the *weakest* contracting direction. If $\lambda_1 > \lambda_2^2$ this is the tangential direction ($\mu > \lambda_2$), so $N_+^1 \partial A$ is a smooth graph and ∂A is genuinely curved at $p_\pm$. If $\lambda_1 < \lambda_2^2$ the weakest direction is ∂_{x_2}, which is normal to the fibre, so $N_+^1 \partial A$ arrives at p_+ tangent to ∂_{x_2}. In that case $\theta \sim |x_2|^{1/(1-\alpha)}$ along $N_+^1 \partial A$, and ∂A is flat to higher order at $p_\pm$. The resonance is the transition between the two regimes, which is where the task sheet expects singularities to be born under the perturbation φ.

3.5 Contact spectral pairing: a linear contact algebra viewpoint

[Draft note: [added] Strand (A) in action. The general statement is standard reasoning but should be checked against the literature on conformally symplectic linear algebra.]

Proposition 3.10 (spectral pairing at fixed points)**.** *Let T be a contactomorphism of a $(2m+1)$-manifold with $T^*\alpha = \rho\,\alpha$, and let $T(z) = z$. Then DT_z preserves $\xi_z = \ker \alpha_z$, acts on T_zM/ξ_z as multiplication by $\rho(z)$, and $DT_z|_{\xi_z}$ is conformally symplectic for $d\alpha|_{\xi_z}$ with factor $\rho(z)$. Hence*

$$\sigma(DT_z) = \{\rho(z)\} \;\cup\; \{\nu_1, \rho/\nu_1, \ldots, \nu_m, \rho/\nu_m\}.$$

Proof. At a fixed point, $\alpha_z(DT_z v) = \rho(z)\alpha_z(v)$, which gives the first two claims. Moreover $T^*d\alpha = d\rho \wedge \alpha + \rho\, d\alpha$, and on ξ_z the first term vanishes. A linear map S with $S^*\omega = \rho\,\omega$ satisfies $S^T \Omega S = \rho\,\Omega$, so S is similar to ρS^{-T} and its spectrum is invariant under $\nu \mapsto \rho/\nu$. □

For the boundary map, $\rho = \|Df(x)^{-T} n\|^{-1}$ (Lemma 2.2). In the diagonal case, $\xi_{p_+} = \operatorname{span}\{\partial_{x_2}, \partial_\theta\}$ and $\rho(p_+) = \|L^{-1} e_1\|^{-1} = \lambda_1$. Indeed $\sigma(J_\beta(p_+)) = \{\lambda_1\} \cup \{\lambda_2, \lambda_1/\lambda_2\}$, and the pair multiplies to ρ. At $q_\pm$, $\rho = \lambda_2$ and the pair is $\{\lambda_1, \lambda_2/\lambda_1\}$. This reorganises Section 3.4:

- The Legendrian tangent direction lies in ξ (Legendrian submanifolds are tangent to $\ker \alpha$), so it carries one member ν_T of a pair. Since orbits approach a sink along the weakest direction, ν_T is the *larger* member of the pair: $\nu_T = \mu$ if $\lambda_1 > \lambda_2^2$ and $\nu_T = \lambda_2$ if $\lambda_1 < \lambda_2^2$.

- The resonance $\lambda_1 = \lambda_2^2$ is precisely the *collision* $\nu = \rho/\nu = \sqrt{\rho}$ within the pair, where the Legendrian tangent switches from one member of the pair to the other. Locally at $p_\pm$ this collision is the only place where normal hyperbolicity fails (Proposition 3.6(i)). The failure on the whole region $\lambda_1 < \lambda_2^2$ under the uniform definition (Proposition 3.6(ii)) is a global mismatch between $p_\pm$ and $q_\pm$.

- By analogy with Krein collisions, a generic perturbation through a collision can send the pair into the complex plane, onto the circle $|\nu| = \sqrt{\rho}$. This is the mechanism suggested by the task sheet for the birth of singularities. We verified the pairing numerically at fixed points of boundary maps of a non-diagonal linear map, the Task 2 map and a Hénon-like map. In each case the eigenvalues on ξ multiply to ρ to machine precision.

3.6 Numerical experiments

[Draft note: [added] Figures and checks generated while drafting with the attached `boundary_map_numerics.py` and `pairing_checks.py` (JAX). Re-run and adapt them yourself.] The boundary map was implemented in JAX in the coordinates (x_1, x_2, θ), with Jacobians computed by automatic differentiation. The following checks were performed:

- the fixed points of Proposition 3.1 satisfy $|\beta(z) - z| < 10^{-15}$, and the spectra and eigenvectors of Proposition 3.3 are reproduced to machine precision for $(\lambda_1, \lambda_2) = (0.3, 0.5)$ and $(0.4, 0.7)$, including the sign of $(-\tilde{R}, 0, 1)$ at q_+;
- $W^u(q_\pm)$ from fundamental-domain iteration agrees with the series of Theorem 2.1(iii) to 3×10^{-15}, confirming Proposition 3.5 (Fig. 3);
- $\beta^* \alpha = \alpha / \|L^{-T} n\|$ at random points (Lemma 2.2);
- at $(\lambda_1, \lambda_2) = (0.49, 0.7)$, $J_\beta(p_+)$ has $\operatorname{rank}(J_\beta - \lambda_2 I) = 2$, i.e. a Jordan block;
- the radius of curvature at p_+ converges to R as $\theta \to 0$ in the normally hyperbolic regime (at $\theta = 10^{-10}$ it is 5.998 versus $R = 6$, with the correction $\asymp \theta^\gamma$), has fitted log-slope -1.4427 (predicted $-1/\log 2$) at the resonance $(0.25, 0.5)$, and has fitted power -0.247 (predicted -0.2435) at $(0.2, 0.5)$ (Fig. 4).

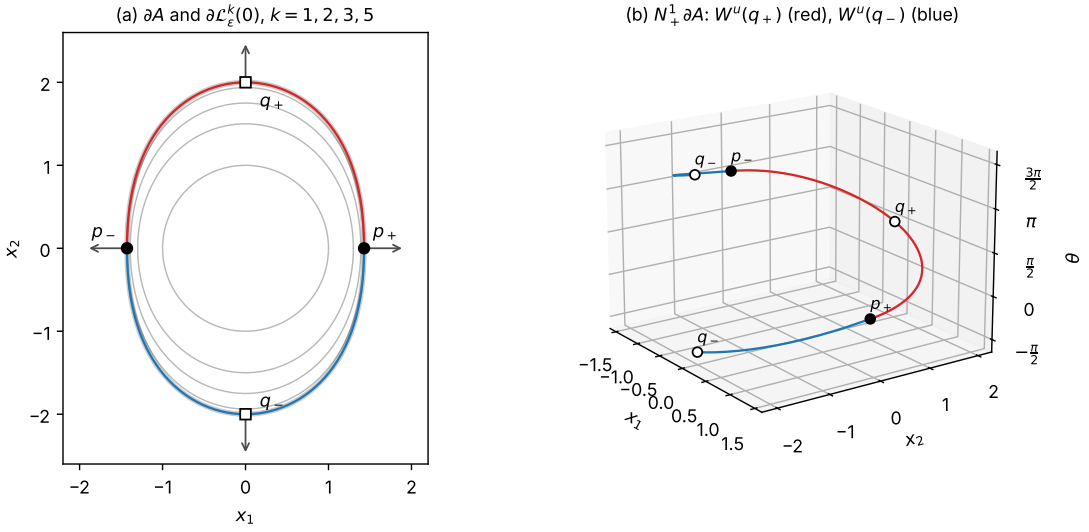


Figure 3: $(\lambda_1, \lambda_2, \varepsilon) = (0.3, 0.5, 1)$. (a) The attractor boundary ∂A (thick grey), with the nested sets $\partial \mathcal{L}^k_\varepsilon(0)$ converging to it, and the projections of $W^u(q_+)$ (red) and $W^u(q_-)$ (blue) computed by fundamental-domain iteration. Arrows are the outward normals at the fixed points. (b) The same curves in $\mathbb{R}^2 \times \mathbb{S}^1$: the Legendrian curve $N^1_+ \partial A$ is the union of the four fixed points and the four heteroclinic branches.

4 Research plan

4.1 Next steps: the resonant perturbation (Tasks 2 and 3)

Fix $\lambda_1 = \lambda^2$, $\lambda_2 = \lambda$ and $f(x) = (\lambda^2 x_1 + a x_2^2,\ \lambda x_2)$. This is a diffeomorphism with $f^{-1}(y) = (\lambda^{-2}(y_1 - a\lambda^{-2} y_2^2), \lambda^{-1} y_2)$ and $Df(0) = L$.

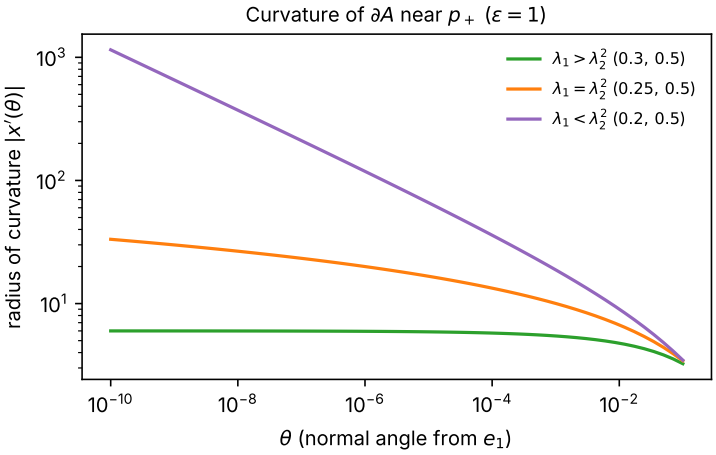


Figure 4: Radius of curvature $|x'(\theta)|$ of ∂A near p_+ in the three regimes of Proposition 3.8: bounded (normally hyperbolic), logarithmic divergence (resonance), power-law divergence.

(a) Fixed points of $\beta_{f,\varepsilon}$. These solve $x - f(x) = \varepsilon n$ and $Df(x)^T n = cn$ with $c > 0$, where

$$Df(x)^{-T} = \begin{pmatrix} \lambda^{-2} & 0 \\ -2ax_2\lambda^{-3} & \lambda^{-1} \end{pmatrix}.$$

[Draft note: The notes use $(L+\varphi)^{-T} = L^{-T} + \varphi^{-T}$, which does not hold. The boundary map needs the derivative $Df(x)^{-T} = (L + D\varphi(x))^{-T}$, as above.] Since $Df(x)^T$ is lower triangular with diagonal (λ^2, λ), either $n = \pm e_2$ ($c = \lambda$; continuations of $q_\pm$, with $x = (ax_2^2/(1-\lambda^2), \pm\varepsilon/(1-\lambda))$), or $c = \lambda^2$ and $n_2 = 2ax_2n_1/(\lambda^2 - \lambda)$. Combined with $x_2 = \varepsilon n_2/(1-\lambda)$, the latter gives $n_2 = 0$ ($p_\pm$, unchanged) or

$$n_1 = -\frac{\lambda(1-\lambda)^2}{2a\varepsilon}, \qquad n_2 = \pm\sqrt{1-n_1^2}, \qquad x_2 = \frac{\varepsilon n_2}{1-\lambda}, \quad x_1 = \frac{\varepsilon n_1 + ax_2^2}{1-\lambda^2}.$$

[Draft note: [added] Derived while drafting and checked numerically (residual $\sim 10^{-17}$).] This pair of extra fixed points exists iff $|a|\varepsilon > \lambda(1-\lambda)^2/2$, and it is related by the reflection of Remark 3.4. Numerically they are sinks; at $\lambda = 0.6$, $\varepsilon = 0.5$, $a = 0.2$ their spectrum is $\{0.36, 0.768, 0.469\}$, and the pair in ξ again multiplies to ρ. Note also that the isotropic rescaling $x \mapsto cx$ sends $(a, \varepsilon) \mapsto (a/c, c\varepsilon)$, so for fixed λ the problem depends only on the product $a\varepsilon$. The bifurcation study is therefore one-parameter, in $a\varepsilon$, and every threshold below is a condition on $a\varepsilon$.

(b) Spectra. The Jacobian is no longer block-triangular (Lemma 3.2 fails because the normal update depends on x). Compute $\sigma(D\beta_{f,\varepsilon})$ at the fixed points and locate the curves in the parameter $a\varepsilon$ where eigenvalues become complex. The task sheet's hint links these to the birth of singularities. A first computation is encouraging.[Draft note: [added] Two-line computation plus numerical check; verify.] The points $p_\pm$ survive unchanged, since $f(x_1, 0) = (\lambda^2 x_1, 0)$, and $\rho(p_\pm) = \lambda^2$. On $\xi_{p_+} = \operatorname{span}\{\partial_{x_2}, \partial_\theta\}$, however, the nonlinearity creates the coupling $\partial\theta'/\partial x_2 = -2a/\lambda$, giving

$$D\beta_{f,\varepsilon}(p_+)|_\xi = \begin{pmatrix} \lambda - 2a\varepsilon/\lambda & \varepsilon\lambda \\ -2a/\lambda & \lambda \end{pmatrix}, \qquad \det = \lambda^2 = \rho, \quad \operatorname{tr} = 2\lambda - \frac{2a\varepsilon}{\lambda}.$$

So the pair at p_+ is complex, on the circle $|\nu| = \lambda = \sqrt{\rho}$, if and only if $0 < a\varepsilon < 2\lambda^2$. At p_- the sign of a flips, giving complex eigenvalues iff $-2\lambda^2 < a\varepsilon < 0$. This is consistent

with Proposition 3.10 and Remark 3.4, and numerically ($\lambda = 0.6$, $a = 0.4$, $\varepsilon = 0.5$) the eigenvalues at p_+ are $0.267 \pm 0.537i$.

On the opposite side, the pair at p_- is real for $a > 0$, with trace $2\lambda + 2a\varepsilon/\lambda$. Its larger member exceeds 1 iff $\mathrm{tr} > 1 + \rho$, i.e. iff $a\varepsilon > \lambda(1-\lambda)^2/2$. This is *the same threshold* at which the extra fixed points of (a) appear, with $n_1 \to -1$ as $a\varepsilon \downarrow \lambda(1-\lambda)^2/2$. This strongly suggests a pitchfork bifurcation at p_- (reflection-symmetric, hence a pitchfork) that turns p_- into a saddle and creates the two new sinks. Numerically, at $\lambda = 0.6$, $\varepsilon = 0.5$, $a = 0.2$ the spectrum at p_- is $\{0.36, 1.244, 0.289\}$. It is a natural candidate for singularities born while the eigenvalues are still *real*, which is the task sheet's second question. The next steps are to relate the focus at p_+ and the saddle at p_- to the geometry of $\mathcal{L}'$ near these points.

(c) The invariant Legendrian curve $\mathcal{L}'$. Construct it numerically as a union of unstable manifolds of saddles of $\beta_{f,\varepsilon}$, using fundamental-domain iteration, and theoretically by a graph transform over $N^1_+\partial A$. Show that $\mathcal{L}' \supseteq \overline{N^1_+\partial A'}$.

(d) Fronts and singularities. Project $\mathcal{L}'$ to the big front $\Gamma' \subset \mathbb{R}^2$. Detect self-intersections (A_1^2), and cusps (A_2, where $\mathcal{L}'$ is tangent to the fibre ∂_θ, i.e. $x'(\theta) = 0$). Show that $\partial A'$ carries only wedges while Γ' also has cusps in the interior of A'. Test the conjecture on the task sheet: the singularities are stable but not isolated, accumulating at the projection of a fixed point with complex eigenvalues. A focus-type fixed point would make $\mathcal{L}'$ spiral, producing infinitely many cusps.

(e) Asymmetry. Following Remark 3.4, compare the two sides of A' for $a > 0$ and $a < 0$.

4.2 Two broader strands

The task sheet is one instance of two more general programmes, which could form the second half of the thesis.

(A) Linear contact algebra. Linear symplectic algebra (the group $Sp(2m)$, its spectral symmetry, normal forms of symplectic matrices, Krein signature and collisions [6]) underpins Hamiltonian dynamics. The analogous *linear contact algebra* concerns the linearisations DT_z of contactomorphisms at fixed and periodic points, equivalently the group of linear maps of $(\mathbb{R}^{2m+1}, \alpha)$ preserving $\ker\alpha$ conformally with induced conformally symplectic action. Questions:

(A1) Normal forms of linear contactomorphisms under linear contact conjugacy. Which Jordan structures are allowed, and what is the analogue of Krein signature for colliding pairs $\nu = \rho/\nu$?

(A2) Which spectral configurations are realised by boundary maps? Here $\rho = \|Df^{-T}n\|^{-1}$ is tied to f, which may constrain e.g. ρ relative to the pair. Classify the codimension-one degenerations (collisions, $|\nu| = 1$, $\nu = \rho$) and match each with a regularity or singularity phenomenon of attractor boundaries. Propositions 3.6 and 3.10 give the first entry of such a dictionary: pair collision at a sink $\Leftrightarrow$ switch of the Legendrian tangent between the members of the pair $\Leftrightarrow$ transition from positive to vanishing curvature of ∂A at $p_\pm$, and local failure of normal hyperbolicity.

(A3) Lyapunov-exponent pairing along $N^1_+\partial A$ [8] and its consequences for C^r regularity of ∂A via r-normal hyperbolicity.

(A4) Higher dimensions: several pairs, and resonances between them.

(B) Pontryagin flows and control theory. By **??**, boundary maps are spherised Pontryagin extremal maps, and in continuous time the boundary dynamics is the contact reduction of the Pontryagin flow of $H = \langle p, F(x)\rangle + \varepsilon\|p\|$. Questions:

(B1) Make the dictionary precise: attractors and minimal invariant sets $\leftrightarrow$ (closures of) reachable sets and control sets [5]; boundaries $\leftrightarrow$ envelopes of extremals; invariant Legendrian submanifolds $\leftrightarrow$ invariant Lagrangian cones of the Pontryagin Hamiltonian system.

(B2) Import results: singularities of reachable sets and optimal syntheses (switching curves, conjugate points, cut loci) [3] should correspond to wedges and cusps of fronts. In particular, cusps of the big front Γ' should be conjugate-point (caustic) phenomena of the extremal flow. This would give a control-theoretic route to Task 3.

(B3) Continuous time: attractors of random differential equations with bounded noise [13] via the Pontryagin flow, with the boundary map recovered as a time-h map.

(B4) Numerics: exploit the extremal formulation (shooting, continuation of extremals) in JAX, alongside the fundamental-domain methods of Section 3.

(B5) Beyond spherical noise: for a general convex control set U, $\varepsilon\|p\|$ becomes the support function $h_U(p)$, which handles the non-spherical noise question of [17] within the same framework.

4.3 Longer-term directions

1. **Regularity of ∂A for general linear L.** Extend Propositions 3.6 and 3.8 to $d \geq 2$ and to non-diagonalisable L or L with complex eigenvalues. In the latter case $L_\perp$ has no fixed points on $\mathbb{S}^1$, and the dynamics on $N^1_+\partial A$ is a circle rotation. Relate r-normal hyperbolicity ($a < b^r$) to C^r regularity of ∂A.

2. **Persistence and bifurcation of boundaries.** Combine the singularity picture with the persistence theory [15] to describe codimension-one bifurcations of attractor boundaries. Connect this with the planar classification of [19, 20].

3. **Nonlinear examples and numerics.** Apply the methods to genuinely nonlinear systems, e.g. the Hénon map with bounded noise [18]. Build a reusable JAX toolbox for boundary maps (automatic Jacobians, invariant-manifold continuation), and compare with ε-chain and set-oriented methods.

4.4 Timeline to the Late Stage Review

[Draft note: Adjust months to your actual start date; the LSR is due by month 24.]

Period	Goal
Months 12–14	Task 2(a,b): fixed points and spectra of $\beta_{f,\varepsilon}$; numerical bifurcation diagram in $a\varepsilon$.
Months 14–17	Task 2(c,d): Legendrian curve and front; numerical evidence of wedges and cusps.
Months 17–20	Task 3: classification and stability of the singularities; proofs in the simplest cases.
Months 20–22	Rigorous version of Propositions 3.6, 3.8 and 3.10 for general planar L; draft paper.
Throughout	Reading for strands (A) and (B): linear symplectic algebra and Krein theory; geometric control and reachable sets. Aim for a precise (A)/(B) research question by the LSR.
Months 22–24	LSR report: paper-level account of the resonant example; thesis outline including strands (A), (B).

Risks. The rigorous singularity analysis (Task 3) may be hard. As a fallback, the thesis can rest on (i) the regularity theory for linear boundaries and (ii) careful numerical exploration with conjectures, which are publishable on their own.

5 Training and development

[Draft note: Fill in from your records.]

- Taught courses (towards the departmental 100-hour requirement): [e.g. LTCC / TCC / MAGIC / Imperial MSc modules, with hours].
- ECRI professional skills: [courses and credits; 2 needed by the ESA].
- Plagiarism Awareness course: [completed / date]. Trusted Research e-learning: [completed / date].
- DACR / Centre for Academic English initial training: [status].
- Seminars and reading groups: [dynamical systems seminar, etc.].
- Self-study: differential geometry of tangent and normal bundles; stable manifold theorem (Perron method); linear algebra and spectral theory; contact geometry and Legendrian fronts.

Use of generative AI

[Draft note: Required by the 2026 ESA guidance (para. 4). Edit to reflect what you actually used and kept.] Claude (Anthropic) was used to transcribe handwritten notes, draft the structure and prose of this report, propose proofs of Lemma 2.2 and **??**, Propositions 3.5, 3.6, 3.8 and 3.10, Remark 3.4 and the computation in Section 4.1(b), and write the numerical script used for Figs. 3 and 4. All mathematical content was checked by the author [describe how: re-derived by hand, numerics re-run, discussed with supervisor], and [describe what was rewritten].

[Draft note: Check bibliographic details; add arXiv numbers where missing.]

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